

B.Tech IV Year I Semester (R20) Regular & Supplementary Examinations December/January 2024/2025

BRIDGE ENGINEERING

(Civil Engineering)

Time: 3 hours

Max. Marks: 70

PART – A

(Compulsory Question)

1 Answer the following: (10 X 02 = 20 Marks)

- | | |
|---|----|
| (a) What are the factors to be considered while selecting suitable site for a bridge? | 2M |
| (b) Define elastomeric pad bearings. | 2M |
| (c) List the types of Culvert. | 2M |
| (d) Discuss the different classes of live loads to be considered in deck slab design of a bridge. | 2M |
| (e) Discuss the advantages of T beam bridges. | 2M |
| (f) What do you understand by racking force and where this term is used? | 2M |
| (g) Draw the sketch of various elements of plate girder. | 2M |
| (h) Discuss the advantages of composite bridges. | 2M |
| (i) State the requirements of good pier. | 2M |
| (j) Discuss any two types of foundation provided for R.C.C. bridges. | 2M |

PART – B

(Answer all the questions: 05 X 10 = 50 Marks)

- 2 Design a RC box culvert having clear vent way of 3 m by 3 m. The dead load is 14 kN/m^2 and live load is 50 kN/m^2 . Unit weight of soil = 18 kN/m^3 . Angle of repose = 30° . Use M25 grade of concrete and Fe415 grade of steel. Also sketch detailing of reinforcements. 10M

OR

- 3 Design an elastomeric pad bearing for a two lane reinforced concrete T-beam bridge for 15 m effective span having the following data: 10M
 Vertical sustained load- 300 kN
 Vertical dynamic load- 60 kN
 Horizontal sustained load- 80 kN
 Coefficient of friction = 0.65
 Modulus of rigidity - 1 N/mm^2
 M 20 grade concrete.
- 4 Design a solid slab bridge required for a highway road having the following data. 10M
 Width of carriage way = 7.5 m
 Clear Span = 5 m
 Loading = IRC Class A
 Width of Kerb = 600 mm
 Materials = M 30 concrete and Fe 415 grade steel.

OR

- 5 Design the cantilever footpath portion of a bridge having a clear overhang of 1600 mm considering live load and most un favourable vehicular load. Consider M20 and Fe 500 combination. Given that width of long girder is 300 mm. 10M

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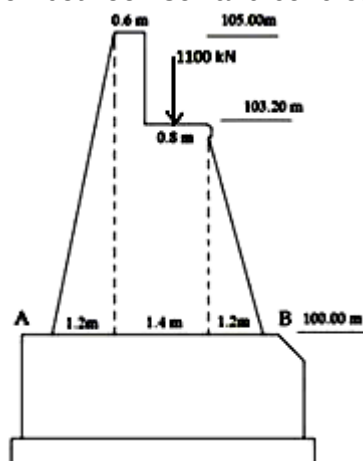
- 6 Design the intermediate longitudinal girder of a T beam and slab bridge for the following data: 10M
 Effective span = 10 m
 Carriage way width = 7.5 m
 Kerb = 600 mm width on either side
 Provide three longitudinal beams.
 Loading = IRC Class A vehicle
 Adopt M30 concrete and Fe 415 grade steel. Shear check is not required.

OR

- 7 What are the load distribution theories used in girders of Tee beam and slabbridge? Explain Courbon's theory. 10M
- 8 Calculate dead load for an intermediate longitudinal girder for the following data. 10M
 Effective span = 21 m Slab thickness = 210 mm Width of rib = 300 mm Size of bottom flange = 300 mm x 600 mm Spacing of main beam = 3 m Overall depth of beam = 1.8 m.

OR

- 9 Design a steel longitudinal girder for a steel concrete composite construction. Thickness of RCC deck slab = 250 mm; effective span = 12.25 m. Centre to centre distance of girders = 2.5 m. Consider appropriate live load. 10M
- 10 Verify the stability of abutment. The salient details are given below. 10M
 Material of the abutment : Concrete
 Live load : IRC AA (Tracked)
 Density = 18 kN/m^3
 Angle of repose = 30°
 Coefficient of friction = 0.6
 Span of bridge = 15 m
 Angle of friction between soil and concrete = 18°



OR

- 11 Describe the different types of bridge piers Sketch a typical cross section of any one pier and indicate the forces acting on it. 10M

B.Tech IV Year I Semester (R19) Supplementary Examinations December 2024

BRIDGE ENGINEERING

(Civil Engineering)

Time: 3 hours

Max. Marks: 70

PART – A

(Compulsory Question)

- 1 Answer the following: (10 X 02 = 20 Marks)
- | | |
|--|----|
| (a) Give some IRC specifications for road bridges. | 2M |
| (b) State the various types loads to be considered in the design of bridges. | 2M |
| (c) Write the types of culverts. | 2M |
| (d) Write the Advantages of Box Culvert. | 2M |
| (e) Draw a neat sketch of component of a T-beam bridge. | 2M |
| (f) Write the uses of stiffeners. | 2M |
| (g) Evaluate the main function of vertical and bearing stiffeners in a plate girder. | 2M |
| (h) Write the advantages of plate girder bridges. | 2M |
| (i) Show different types of piers used for bridges. | 2M |
| (j) List the various loads that are to be considered for a pier design. | 2M |

PART – B

(Answer all the questions: 05 X 10 = 50 Marks)

- 2 Design an elastomeric pad bearing for a two lane reinforced concrete T-beam bridge for 15 m effective span having the following data: 10M
- Vertical sustained load- 300 kN
 Vertical dynamic load- 60 kN
 Horizontal sustained load- 80 kN
 Coefficient of friction = 0.65
 Modulus of rigidity-1 N/mm²
 M 20 grade concrete.

OR

- 3 (a) Write about the site investigation in bridge design. 5M
 (b) Explain about the highway bridge loading standards. 5M
- 4 Design a box Culvert having inside dimensions of 3 m x 3 m. This culvert is subjected to a dead load of 14,000 N/m² and a Live load of IRC Class AA tracked Vehicle. Assume Unit Weight of soil to be 18,000 N/m³ the angle of repose of soil is 30 degree .Use M25 Concrete and F415 steel Road width is 7.5 span is 3.3 m. 10M

OR

- 5 A reinforced concrete simply supported slab forms the deck of a road bridge, having the following data: 10M
- Clear span = 6 m. : Carriage way -2 Lane. : Width of kerb = 500 mm on either side: Width of bearing = 400 mm: Materials = M25 grade concrete and Fe 415 steel. : Type of loading IRC class AA tracked vehicle.
 Design the deck slab. Show the reinforcement details.

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- 6 Design and report a Tee beam for a bridge deck required for the crossing of highway using the following data. Clear width of roadway = 12 m, Effective span of the bridge = 21 m, Loading = IRC Class AA Number of main girders = 7, Spacing of the main girders = 2 m, Spacing of cross girders = 3 m, Materials = M30 grade concrete and Fe415 steel. 10M
- OR**
- 7 Design and report a Tee beam for a bridge deck required for the crossing of highway using the following data. Clear width of roadway = 15 m, Effective span of the bridge = 22 m, Loading = IRC Class AA Number of main girders = 8, Spacing of the main girders = 2 m, Spacing of cross girders = 4 m, Materials = M30 grade concrete and Fe415 steel. 10M
- 8 Solve a welded plate girder for a B.G track using the following data Effective span of the plate girder = 20 m Dead load of the track = 8 kN/m EUDL of live load on track for Bending Moment = 2000 kN EUDL of live load on track for shear force = 2000 kN. 10M
- OR**
- 9 A plate girder is to be designed for a B.G track to suit the following data. Span of the bridge = 20 m, Dead load of the track = 7.5 kN/m EULL for B.M calculation / track = 2027 kN EULL for shear calculation / track = 2224 kN Solve for the plate girder alone. Stiffeners and bracings need not to be designed. 10M
- 10 (a) List the design loads to be considered in abutments? And discuss about wing walls. 5M
(b) Write about the general features of a Abutment. 5M
- OR**
- 11 (a) Distinguish between Pier and Abutment with reference to Bridge substructure. 5M
(b) Explain the consideration in selecting the orientation of wing walls in the design of bridge abutments. 5M

B.Tech IV Year I Semester (R15) Supplementary Examinations December 2024

BRIDGE ENGINEERING

(Civil Engineering)

Time: 3 hours

Max. Marks: 70

PART – A
(Compulsory Question)

- 1 Answer the following: (10 X 02 = 20 Marks)
- Define Impact factor.
 - Mention the classification of bridges.
 - Define clearance and width of carriage way.
 - List out the various IRC codes to be used while designing the deck slab bridge for a national highway.
 - Differentiate between straight slab bridge and skew slab bridge.
 - State the advantages of Pigeauds method.
 - What is composite action in a composite girder bridge?
 - Mention the elements of a plate girder
 - Define Dirt wall and wing wall.
 - Mention the different types of piers used bridges.

PART – B

(Answer all the questions: 05 X 10 = 50 Marks)

- 2 (a) Explain the components of a bridge structure with neat diagram.
(b) Mention the factors to be considered while selecting suitable site for a bridge.
- OR**
- 3 Design an elastomeric pad bearing for a two lane reinforced concrete T-beam bridge for 15 m effective span having the following data:
Vertical sustained load- 310 kN
Vertical dynamic load- 50 kN
Horizontal sustained load - 80 kN
Coefficient of friction = 0.65
Modulus of rigidity-1 N/mm².
- 4 A reinforced concrete simply supported slab forms the deck of a road bridge, having the following data:
(i) Clear span = 6 m.
(ii) Carriage way -2 Lane.
(iii) Width of kerb= 500 mm on either side.
(iv) Width of bearing= 400 mm.
(v) Materials = M25 grade concrete and Fe 415 steel.
(vi) Type of loading IRC class AA tracked vehicle.
Design the deck slab.
- OR**
- 5 Design a deck slab for the following details:
Carriage way-two lane (7.5 m width), Foot paths – 1 m on either side, Clear span - 6 m, Width of wearing coat – 80 mm, width of bearing – 400 mm, Materials: M25 grade concrete and Fe415 steel HYSD bars, Loading: class AA tracked vehicle.

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- 6 Design an intermediate T-Beam for the following data- slab need not be designed. (i) Road width 8.00 m, Total width = 10.68 m, (ii) Span 20.00 m, (iii) Number of main girders 4 and cross girders 6, (iv) Concrete Grade M40, (v) Thickness of slab 250 mm, (vi) Steel Fe 500, (vii) Thickness of wearing coat 75 mm, (viii) Loading Class AA tracked. No reinforcement detailing is required.

OR

- 7 Using pigeauds curve and charts, Design an interior slab of a T beam bridge measuring 5 m × 6 m for an IRC class A loading with M30 concrete and Fe 500 steel.

- 8 In a composite bridge, the cube compressive strength of concrete in the slab is found to be 35 MPa. Determine the static shear strength of the following shear connectors embedded in the slab. (i) Stud (Diameter 18 mm and height 80 mm), (ii) Channel ISMC 125 (Flange width 65 mm, Depth = 125 mm) of length 200 mm.

OR

- 9 What is the design procedure for plate girder bridges? Explain step by step.

- 10 (a) What are the types of foundations in bridges? Explain any one in detail with neat sketches.
(b) Discuss about the stability analysis of abutments.

OR

- 11 (a) Explain using sketches, the different safety checks made for piers of bridges.
(b) Enumerate the forces to be taken into account while designing an abutment of a bridge.
